

PROFESSIONAL EXPERIENCE

Nuclear Software Engineer at X-energy [November 2025 - Present]

- *XSTERM*: C# development using **.NET 8**
 - Implementation of **Monte Carlo** simulation methods to perform uncertainty quantification analysis of reactor and fuel models.
 - Implementation of a performance monitoring library for use in a **Gitlab pipeline** to track performance degradations across various simulation runs in every merge request.
 - Development of unit, integration and system tests using **NUnit**. Reconciled existing unfinished test suite with software requirements to fill gaps in testing.

Radar Software Engineer at C Speed [March 2025 - November 2025]

- *Light Wave Radar*: C and C++ development for a **RTOS**
 - Creation of a **SQLite** database interface which receives, organizes and stores high frequency data streams from multiple sources.
 - Addition of a **TCP** socket communication layer to multiple applications enabling bidirectional communication between multiple machines on a **LAN**
 - Development of test server **powershell** scripts used to run data processing software in a vacuum and check output using binary comparisons. Automated this process using **Gitlab pipelines**.

Software Engineer Consultant at C Speed [June 2023 - March 2025]

- *Patient Monitor Medical Device*: C and C++ development for a **Yocto Linux** embedded OS
 - **Linux kernel** development including removal/modification of various drivers, creation of udev rules, and adjusting kernel configs
 - Application UI overhaul using C++ with **Qt** graphical framework adhering to FDA usability requirements and **UI/UX** design standards provided by the client
 - Development of an automated **bitbake** build process running on an **Azure** cloud **Ubuntu** VM using **Jenkins** continuous integration
- *Medical Device Service Tool*: C# development using **.NET** framework 3.5
 - Development of a **WPF** desktop application which interfaces with medical devices via USB connection
 - Implementation of software security features such as dynamic **AES encryption**, password complexity policies, and automatic timeout/logout

Student Programmer at Seton Hill University [June 2022 - June 2023]

- *Student/Faculty Web Portal*: Migration of a **PHP** web app from Symfony to **Laravel** framework
 - Migration of various student and faculty facing features using **PHP** with Laravel/Filament to query and present data from a **MySQL** database

Software Engineer Intern at C Speed [May 2022 - August 2022]

- *Borescope Device*: C# development for an embedded **Windows** platform
 - UI updates, bug fixes, and unit tests for borescope device engine inspection features

TECHNICAL SKILLS

Programming Languages: C++, C#, C, Python, Swift, Javascript, PHP, Bash, Powershell

Tools: Git, Tortoise SVN, Azure, TFS, Visual Studio, VSS, WSL, Jira, GCC, CMake, Yocto, Uboot, Bitbake, SSH, PSOC, Tera Term, .NET, React, Qt, Jenkins, Docker, VirtualBox, Vercel, SQLite, NUnit, GoogleTest

OPEN SOURCE CONTRIBUTIONS

MOOSE: Multiphysics Object Oriented Simulation Environment

Idaho National Laboratory [<https://github.com/idaholab/moose>]

Cardinal: High-Fidelity Multiphysics Simulation

Argonne National Laboratory [<https://github.com/neams-th-coe/cardinal>]